

MARREE, Australia

WMO: 954800

Lat: 29.659S

Lon: 138.068E

Elev: 166

StdP: 14.61

Time Zone: 9.50 (AUA)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	36.2	38.6	10.9	9.7	72.8	14.4	11.5	75.5	27.4	65.5	24.3	63.5	7.2	150	0.563

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	26.2	109.7	68.2	106.8	67.4	103.9	66.8	75.4	87.5	73.3	90.1	71.6	91.2	13.1	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.0	123.3	78.2	69.2	107.9	77.7	65.9	95.9	78.4	39.0	84.8	37.0	89.8	35.4	91.2	82.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 29.1	25.3	22.5	DB	32.0	113.6	1.9	2.8	30.6	115.6	29.5	117.2	28.5	118.8	27.1	120.8	(4)
(5)			WB	28.4	77.9	2.1	2.7	26.8	79.8	25.6	81.4	24.4	82.9	22.9	84.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	70.4	87.4	84.9	78.9	71.1	61.3	54.6	54.1	57.1	64.5	71.8	78.0	82.6		
	DBStd	13.15	7.14	7.25	6.70	6.20	5.47	4.35	4.66	5.27	6.63	7.08	7.47	7.35		
	HDD50	20	0	0	0	0	0	8	10	2	0	0	0	0		
	HDD65	1175	0	0	1	16	144	313	340	254	89	17	1	0		
	CDD50	7477	1160	976	896	632	349	146	137	223	435	675	839	1009		
	CDD65	3158	695	556	432	198	28	1	2	10	75	226	391	544		
	CDH74	46005	11020	8295	5679	2272	330	13	39	196	1187	3233	5546	8195		
	CDH80	27506	7414	5331	3184	967	64	1	2	41	479	1636	3218	5169		
(14) Wind	WSAvg	11.3	12.9	12.2	11.4	10.2	9.3	9.1	9.2	10.5	11.7	12.4	13.0	13.2		
(15) Precipitation	PrecAvg	6.7	1.0	1.0	0.4	0.4	0.6	0.4	0.5	0.4	0.4	0.5	0.4	0.6		
	PrecMax	15.3	7.3	8.0	7.0	3.4	2.0	2.8	2.1	2.4	2.3	2.4	1.8	3.2		
	PrecMin	1.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	3.7	1.5	1.8	1.1	0.8	0.6	0.6	0.6	0.6	0.6	0.6	0.5	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	114.4	111.5	104.5	96.9	85.3	75.8	78.7	84.9	95.0	101.6	108.8	111.5		
		MCWB	69.1	70.2	66.8	64.2	60.2	57.7	57.7	57.1	60.8	63.4	66.1	67.4		
	2%	DB	110.6	107.8	101.1	93.0	81.7	71.3	73.7	79.0	90.2	97.9	103.8	107.8		
		MCWB	68.5	68.8	65.4	63.1	59.5	55.8	54.6	55.2	58.4	61.9	64.7	66.6		
	5%	DB	107.3	104.6	97.8	89.5	77.7	68.1	69.7	75.0	85.6	93.7	99.5	104.1		
		MCWB	68.0	67.5	64.9	61.7	57.6	54.1	53.3	54.1	56.9	60.4	63.5	65.6		
	10%	DB	104.2	101.2	94.5	85.9	73.9	65.5	66.3	71.2	81.0	89.4	95.5	100.1		
		MCWB	67.7	67.3	64.7	60.6	56.1	52.9	52.1	52.5	55.8	59.2	62.6	65.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	78.5	77.5	76.1	70.7	63.9	62.3	62.1	60.6	65.4	67.8	72.4	76.9		
		MCDB	86.2	85.3	81.7	78.7	75.5	66.6	67.6	73.4	77.4	84.4	88.6	83.9		
	2%	WB	75.1	75.2	72.9	67.6	61.4	59.2	56.8	57.9	61.3	64.8	70.0	72.8		
		MCDB	87.4	90.5	83.8	78.9	74.0	65.4	68.3	73.4	81.0	85.9	86.0	88.2		
	5%	WB	72.8	73.2	70.4	64.9	59.6	56.7	54.4	55.5	59.1	62.8	67.7	70.7		
		MCDB	93.9	91.4	86.7	81.8	72.7	63.9	67.6	71.7	80.5	87.5	86.1	89.3		
	10%	WB	71.0	70.7	67.8	62.6	57.8	54.7	52.6	53.5	57.0	61.1	65.5	68.3		
		MCDB	95.9	91.5	88.4	83.3	71.3	63.2	64.9	69.5	77.9	85.5	88.5	92.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	26.2	25.2	25.0	25.3	23.7	22.5	24.7	26.3	27.6	27.8	26.9	26.5		
		MCDBR	29.5	28.0	29.2	29.0	27.2	25.4	28.8	31.3	34.1	33.4	31.3	30.9		
		MCWBR	10.4	9.4	11.1	12.0	12.9	13.9	15.5	15.8	15.4	13.4	12.2	11.2		
	5% WB	MCDBR	22.4	21.1	20.7	21.7	22.3	19.1	25.0	26.6	27.5	26.7	24.0	23.5		
		MCWBR	9.8	8.4	8.7	9.8	11.5	11.4	14.2	14.4	13.2	12.0	10.5	10.1		
(40) Clear-Sky Solar Irradiance	taub		0.366	0.357	0.336	0.325	0.306	0.292	0.297	0.315	0.355	0.356	0.364	0.356		
	taud		2.379	2.435	2.467	2.476	2.496	2.525	2.497	2.429	2.317	2.329	2.349	2.388		
	Ebn at Noon		309	306	303	290	281	278	281	289	291	303	308	313		
	Edh at Noon		41	38	34	31	27	25	27	32	39	42	42	41		
(44) All-Sky Solar Radiation	RadAvg		2559	2340	2047	1593	1224	1052	1147	1466	1875	2227	2437	2565		
	RadStd		118	156	113	65	64	60	45	56	91	117	115	113		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (2 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page